



New fuzzy zeroing neural network with noise suppression capability for time-varying linear equation solving

Dongsheng Guo¹ · Chan Zhang¹ · Naimeng Cang¹ · Xiyuan Zhang¹ · Lin Xiao² · Zhongbo Sun³

Accepted: 6 November 2024 / Published online: 5 February 2025
© The Author(s) 2025

Abstract

Recently, the zeroing neural network (ZNN) with continuous/discrete-time forms has realized success in solving the time-varying linear equation (TVLE). In this paper, we provide a further investigation by proposing a new fuzzy zeroing neural network (FZNN) model to solve the TVLE in noisy environment. Such a FZNN model, which has the capability of suppressing noise, is developed by using the integration enhancement and fuzzy control strategy. Then, theoretical analysis is presented to show that the proposed FZNN model can effectively solve the TVLE, even with the existence of noise. Comparative simulation results through different examples further verify the effectiveness and robustness of the proposed FZNN model on TVLE solving.

Keywords Zeroing neural network (ZNN) · Noise suppression · Time-varying linear equations (TVLE) · Integration enhancement · Fuzzy control strategy

1 Introduction

Time-varying linear equation (TVLE) is of importance in science and engineering field, such as signal processing and robotics (Patrick and Alle-Jan 1998; Hopfield and Tank 1985; Zhang and Guo 2015; Zhang and Yi 2011; Oppenheim et al. 1997). As a result, a variety of methods, (e.g. numerical method Byrd and Waltz 2011), for solving TVLEs have emerged. In 1982, Hopfield proposed a feedback recurrent neural network (Hopfield and Tank 1985; Tank and Hopfield 1986), also known as Recurrent Neural Network (RNN), based on the principles of neurobiology in order to solve the combinatorial optimization problem. RNN

✉ Naimeng Cang
nmcang@163.com

¹ School of Information and Communication Engineering, Hainan University, Haikou, Hainan, China

² College of Information Science and Engineering, Hunan Normal University, Changsha, China

³ Department of Control Engineering, Changchun University of Technology, Changchun 130012, China

is now widely used in numerous fields, including machine translation (Ding et al. 2023), text recognition (Frinken and Uchida 2015), and biomedicine (Shu et al. 2021; Supratak et al. 2017; Xie et al. 2018).

However, traditional gradient-based neural network methods could solve time-invariant equalities and inequalities (Zhang et al. 2009a; Yi et al. 2011; Chen et al. 2013), but are often inefficient in solving time-varying problems and operate with inaccurate results. Therefore, Zhang et al. (2002) proposed a recurrent neural network for solving the time-varying Sylvester equation. Especially, Zhang et al. (2008, 2009b) constructed a classical zeroing neural network (ZNN) with global exponential convergence performance for solving the TVLEs, and also demonstrate the robustness and convergence of the ZNN model by theoretical analyses and simulations. ZNN models with implicit dynamics have superior performance comparing to GNN model in accordance with Zhang et al. (2011a). Due to its superior performance, many scholars keep carrying out researches on ZNNs with various superior performances and have made rapid development since then. Afterwards, Zhang et al. (2013, 2020, 2011b) and Qiu et al. (2018) built discrete ZNN models with various difference formulations, laying the foundation for its successful implementation on hardware.

To speed up the convergence of the ZNN model, Xiao et al. (2018) developed a dynamic neural model to achieve rapid convergence (especially limited time convergence) and noise suppression in ZNNs. However, the limited time convergence is related to the initial state of the system. Research (Xiao et al. 2020) utilizes different activation functions in order to achieve finite time convergence and predefined time convergence of the model. These papers (Zhang and Yan 2018; Zhang et al. 2018) have studied a variety of variable-parameter recurrent neural networks with superior exponential convergence and applied them to the robotic manipulator. As seen above, ZNN models with different superior performances through the introduction of various improving methods.

Noise is an inevitable factor in actual application. Harmonic noise is the representative. Although the ZNN and its variants have had some success in solving TVLEs, it does not take the effects of harmonic noise into account (Zhang et al. 2008, 2009b, 2010, 2011b, 2013, 2019, 2020; Zhang and Guo 2015; Zhang and Yi 2011; Qiu et al. 2018; Lv et al. 2019). Therefore, it is necessary to invest in the model's noise robustness. Jin et al. (2016) enabled the model to be inherently noise tolerant by incorporating an integral term into the model. However, this model could not adjust its convergence rate according to the computational error.

But as the complexity of the system increases, less accurate information is available and the ability to control the system accurately is diminished. Due to this reason, fuzzy control theory has been introduced by Zadeh (1965). Since then, fuzzy control has been widely adopted in the control field, including industry (Precup and Hellendoorn 2011), financial (Cao 2012), water management (Roy and Bhaumik 2018) and so on. Fuzzy control does not depend on its mathematical model of the control system, but on fuzzy rules. Zhang and Yan (2019) proposed an adaptive fuzzy recurrent neural network to effectively reduce joint angle drift and end-effector position accumulation error in the nonrepetitive motion of redundant robotic arms. And to adaptively eliminate the impact of noise, a modified ZNN model is proposed by Jia et al. (2023) on the basis of fuzzy logic control rules.

It is noting that TVLE plays an important role in robotics. Robot arm motion planning can be completed by adaptively compensating the effect of noise and further accurately solving the TVLE. In this paper, inspired by previous successful work, we provide a further

investigation by proposing a new fuzzy zeroing neural network (FZNN) model with noise suppression capability to solve the TVLE in noisy environment. Such a FZNN model exhibits superior robustness and adaptivity. The primary contributions are presented as follows.

- (1) In this paper, a typical fuzzy control strategy is established on the basis of the bipolar sigmoid membership function and nine *if-then* fuzzy logic control rules. Such a strategy differs from the existing strategy used for ZNN.
- (2) In this paper, by using the above fuzzy control strategy and the integration enhancement, the new FZNN model is proposed and studied to solve the TVLE polluted by noise. Such a FZNN model has yet been reported before.
- (3) In this paper, theoretical analysis and simulation results are provided to verify the effectiveness and robustness of the proposed FZNN model in different noise scenarios.

The remainder of this paper is organized as follows. Section 2 presents the preliminary results on TVLE solving. Section 3 describes the formulation of the proposed FZNN model, and the details of the fuzzy control strategy. Section 4 theoretically analyzes the convergence performance of the proposed FZNN model. Section 5 illustrates the simulation results by using the proposed FZNN model under different noises and coefficients. Section 6 provides a clear overview of the paper and an outlook for future research interests.

2 Time-varying linear equation solving

This part introduces the problem of solving the TVLE, and the detailed formulation of the existing ZNN (OZNN) models.

2.1 Problem statement

In this paper, we consider the online solution of the following TVLE:

$$P(t)y(t) = Q(t), \quad (1)$$

where coefficients $P(t) \in \mathbb{R}^{n \times n}$ is time-dependent as well as full rank, and matrix $Q(t) \in \mathbb{R}^n$ is time-dependent also smooth. $y(t) \in \mathbb{R}^n$ is the vector variable which needs determining. The aim of this paper focuses mainly on the situation where coefficient matrix $P(t)$ is positively definite for $t \geq 0$ to guarantee the existence of the theoretical solution.

2.2 OZNN model

To solve the TVLE (1), the following computation error is defined:

$$e(t) = P(t)y(t) - Q(t) \in \mathbb{R}^n.$$

To make the error $e(t)$ converge to zero, by using the exponential decay formula $\dot{e}(t) = -\eta\phi(e(t))$, the original ZNN (OZNN) model is presented as follows:

$$P(t)\dot{y}(t) = -\dot{P}(t)y(t) + \dot{Q}(t) - \eta\phi(P(t)y(t) - Q(t)), \quad (2)$$

where $\dot{y}(t) \in \mathbb{R}^n$, $\dot{P}(t) \in \mathbb{R}^{n \times n}$, and $\dot{Q}(t) \in \mathbb{R}^n$ correspond to time derivatives of $y(t)$, $P(t)$, and $Q(t)$. $\eta > 0 \in \mathbb{R}$ is a parameter which influences the OZNN convergence rate directly, and $\phi(\cdot) : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is an activated functions array. Specifically, the OZNN model (2) with an array of linear activated functions is transformed into

$$P(t)\dot{y}(t) = -\dot{P}(t)y(t) + \dot{Q}(t) - \eta(P(t)y(t) - Q(t)). \quad (3)$$

It follows from Zhang et al. (2009b) that the simplified OZNN model (3) possesses the global and exponential convergence. By choosing different activation functions, the OZNN model (2) can exhibit different properties, involving finite-time convergence and predefined-time convergence (Xiao et al. 2018, 2020).

2.3 IZNN model

Noise is an unavoidable factor, including differential error and so on, which cannot be ignored while applying the OZNN model (3) in practice. Therefore, it is necessary to study the ZNN model with noise suppression capability.

Referencing to Jin et al. (2016), through the introduction of an integral term into (3) to compensate the noise impact, Jin et al. designed the improved ZNN (IZNN) model as follows:

$$\begin{aligned} P(t)\dot{y}(t) = & -\dot{P}(t)y(t) + \dot{Q}(t) - \eta(P(t)y(t) - Q(t)) \\ & - \gamma \int_0^t (P(\tau)y(\tau) - Q(\tau))d\tau + \vartheta(t), \end{aligned} \quad (4)$$

where $\vartheta(t) \in \mathbb{R}^n$ represents additive noise, $\tau \in \mathbb{R}$ is the integral variable, and $\gamma > 0 \in \mathbb{R}$ is a tuning factor that also impacts the IZNN convergence rate. It is worthwhile noting that

the integral item $\int_0^t (P(\tau)y(\tau) - Q(\tau))d\tau$ makes the IZNN model (4) capable of resisting noises.

3 New FZNN model

In this part, with the introduction of fuzzy control strategy into the IZNN model (4), the new FZNN model is proposed to solve the TVLE (1) in noisy environment.

3.1 Fuzzy control strategy

This subsection presents the fuzzy control strategy that is introduced into (4) for constructing the new FZNN model. Such a strategy is established by a bipolar sigmoid membership function and nine *if-then* fuzzy logic control rules.

In general, the fuzzy logic controller consists of three core units, including fuzzification module (D/F), approximate inference module and clarification module (F/D). Consider $e(t)$

and its derivative $ec = de(t)/dt$ as inputs to the fuzzy logic controller, μ and φ as the fuzzy logic outputs.

- (1) *Fuzzification*. For the first step, the inputs are mapped into fuzzy values and membership function by a fuzzification transformation so as to enable them to be fed into the fuzzy inference module. In this paper, a bipolar sigmoid membership function is applied to fuzzy logic controller.
- (2) *Approximate Inference*. For the second step (being the core step), the fuzzy logic controller converts fuzzy input subset into fuzzy output subset by utilizing fuzzy logic rules.
- (3) *Clarification*. For the third step, the fuzzy logic values obtained from the approximate inference module are translated into specific control signals or decision results to drive the actuator for controlling. The defuzzification method used in this paper is the centroid method.

3.2 Model formulation

By introducing the aforementioned fuzzy control strategy into the IZNN model (4), the following FZNN model with noise suppression capability is thus proposed in this paper to solve the TVLE (1):

$$\begin{aligned} P(t)\dot{y}(t) = & -\dot{P}(t)y(t) + \dot{Q}(t) - (\eta + \mu)(P(t)y(t) - Q(t)) \\ & - (\gamma + \varphi) \int_0^t (P(\tau)y(\tau) - Q(\tau))d\tau + \vartheta(t), \end{aligned} \quad (5)$$

with μ and φ being the fuzzy control values which resizes their values adaptively according to feedback error (i.e., computation error) and fuzzy logic control rules.

4 FZNN convergence analysis

This part provides the theoretical analysis for the convergence performance of the proposed FZNN model (5) both in noise-free and noisy situations.

Theorem 1 In considering the case where a positive definite coefficient matrix is given for the TVLE (1), when the initial state $y(0)$ is a stochastic matrix of variables, the proposed FZNN model (5) exhibits stability even in a random initial state.

Proof Applying the Laplace transformation to FZNN model (5) on the basis of $e(t) = P(t)y(t) - Q(t)$ yields,

$$se_{ij}(s) - e_{ij}(0) = -(\eta + \mu)e_{ij}(s) - \frac{\gamma + \varphi}{s}e_{ij}(s) + \vartheta_{ij}(s), \quad (6)$$

Then the transfer function is derived,

$$e_{ij}(s) = \frac{s(e_{ij}(0) + \vartheta_{ij}(s))}{s^2 + (\eta + \mu)s + (\gamma + \varphi)}. \quad (7)$$

It is clear that the poles of the transfer function (7) are

$$s_1 = \frac{-(\eta + \mu) + \sqrt{(\eta + \mu)^2 - 4(\gamma + \varphi)}}{2}, s_2 = \frac{-(\eta + \mu) - \sqrt{(\eta + \mu)^2 - 4(\gamma + \varphi)}}{2}.$$

It is also known that η, μ, γ and φ are positive. For any η, μ, γ and φ all ≥ 0 , the polar points both s_1 and s_2 are ≤ 0 . Then, the response of system (6) will tend to zero. That is to say, the proposed FZNN model (5) possesses stability. Up to here, the stability of the FZNN model (5) has been proven. $\square$

Theorem 2 When a random initial state $y(0)$ is given, the proposed FZNN model (5) will exhibit convergence when no noise involved.

Proof For no noise, i.e., $\vartheta(t) = 0$, corresponding Laplace transformation for FZNN (5) is,

$$se_{ij}(s) - e_{ij}(0) = -(\eta + \mu)e_{ij}(s) - \frac{\gamma + \varphi}{s}e_{ij}(s), \quad (8)$$

which is further reformulated as below:

$$e_{ij}(s) = \frac{se_{ij}(0)}{s^2 + (\eta + \mu)s + (\gamma + \varphi)}. \quad (9)$$

Moreover, applying the final value theorem (Oppenheim et al. 1997) of the Laplace transformation to (9) yields

$$*\lim_{t \rightarrow \infty} e_{ij}(t) = \lim_{s \rightarrow 0} se_{ij}(s) = \lim_{s \rightarrow 0} \frac{s^2 e_{ij}(0)}{s^2 + (\eta + \mu)s + (\gamma + \varphi)} = 0.$$

That is, error tends to zero over time. The convergence property of the FZNN model (5) in noiseless environment is proved right now. $\square$

When the noise is involved in solving the TVLE (1), the following subsystems of the proposed FZNN model (5) are obtained using Laplace transforms (6).

Theorem 3 A positive definite coefficient matrix is introduced to the TVLE (1). Such a FZNN model (5) will exhibit the ability to converge globally to the theoretical solution for $y(t) \in R^n$ with a random initial state $y(0)$.

Proof As the TVLE (1) involves constant noise, the transfer function of (5) is expressed by

$$e_{ij}(s) = \frac{s(e_{ij}(0) + \vartheta_{ij}/s)}{s^2 + (\eta + \mu)s + (\gamma + \varphi)}, \quad (10)$$

with $\vartheta_{ij}(s) = \vartheta_{ij}/s$ as $\vartheta_{ij}(t) = \vartheta_{ij}$. By the final value theorem (Oppenheim et al. 1997) for Laplace transform,

$$*\lim_{t \rightarrow \infty} e_{ij}(t) = \lim_{s \rightarrow 0} s e_{ij}(s) = \lim_{s \rightarrow 0} \frac{s^2(e_{ij}(0) + \vartheta_{ij}/s)}{s^2 + (\eta + \mu)s + (\gamma + \varphi)} = 0.$$

Right after that, $\lim_{t \rightarrow \infty} e(t) = 0$ is inferred. The proof is ended. $\square$

Theorem 4 When equation (1) is considered to be given a positive definite coefficient matrix $P(t)$ and boundary noise, $y(t) \in R^n$ with a random initial state $y(0)$ will converge to its theoretical solution globally. It means that the FZNN (5) is still able to achieve stable convergence globally with boundary noise.

Proof Since the Laplace transform is different as the noise is in sine or cosine form, It will be discussed individually in the follow-up.

(1) *Sinusoidal noise*: When the noise is in sinusoidal form, its Laplace transformation derives from

$$e_{ij}(s) = \frac{s(e_{ij}(0) + \vartheta_{ij}/(s^2 + \vartheta_{ij}^2))}{s^2 + (\eta + \mu)s + (\gamma + \varphi)}, \quad (11)$$

with $\vartheta_{ij}(s) = \vartheta_{ij}/(s^2 + \vartheta_{ij}^2)$ being the Laplace transformation of $\vartheta_{ij}(t) = \sin(\vartheta_{ij}t)$. Applying the terminal value theorem to (11) yields

$$\lim_{t \rightarrow \infty} e_{ij}(t) = \lim_{s \rightarrow 0} \frac{s^2(e_{ij}(0) + \vartheta_{ij}/(s^2 + \vartheta_{ij}^2))}{s^2 + (\eta + \mu)s + (\gamma + \varphi)} = 0.$$

(2) *Cosine noise*: When the noise is in cosine form, it is possible to obtain

$$e_{ij}(s) = \frac{s(e_{ij}(0) + s/(s^2 + \vartheta_{ij}^2))}{s^2 + (\eta + \mu)s + (\gamma + \varphi)}, \quad (12)$$

with $\vartheta_{ij}(s) = s/(s^2 + \vartheta_{ij}^2)$ corresponding to the Laplace transformation of $\vartheta_{ij}(t) = \cos(\vartheta_{ij}t)$. Applying the final value theorem to (12) yields

$$\lim_{t \rightarrow \infty} e_{ij}(t) = \lim_{s \rightarrow 0} \frac{s^2(e_{ij}(0) + s/(s^2 + \vartheta_{ij}^2))}{s^2 + (\eta + \mu)s + (\gamma + \varphi)} = 0.$$

Then $\lim_{t \rightarrow \infty} e(t) = 0$ can be derived from the above discussions. The convergence of the FZNN model (5) in bounded noise environment are proved. $\square$

Theorem 5 Giving the TVLE (1) a positive definite coefficient matrix $P(t)$, when the system (6) is in boundless noise environmen, $y(t) \in R^n$ will converge stably to its theoretical solution globally with a random initial state $y(0)$. This means that in the absence of external noise, the FZNN (5) is able to achieve convergence property globally.

Proof Similarly the Laplace transform of the subsystem yields the following transfer function as the TVLE (1) is in the boundless noise environment,

$$e_{ij}(s) = \frac{s(e_{ij}(0) + \vartheta_{ij}/s^2)}{s^2 + (\eta + \mu)s + (\gamma + \varphi)}, \quad (13)$$

with ϑ_{ij}/s^2 corresponding to the Laplace transformation of $\vartheta(t)$ as $\vartheta(t)$ is linear noise. Applying the final value theorem (Oppenheim et al. 1997) of Laplace transform to (12) yields,

$$\lim_{t \rightarrow \infty} e_{ij}(t) = \lim_{s \rightarrow 0} s e_{ij}(s) = \lim_{s \rightarrow 0} \frac{s^2(e_{ij}(0) + \vartheta_{ij}/s^2)}{s^2 + (\eta + \mu)s + (\gamma + \varphi)} = \frac{\vartheta_{ij}}{\gamma + \varphi}.$$

From the above equation, it is easy to conclude that $\lim_{t \rightarrow \infty} e(t) = 0$ as $\gamma + \varphi \rightarrow \infty$. The proof is ended. $\square$

In brief, the FZNN model (5), regardless of which noise environment, possesses stability and global convergence.

5 Simulation verification

In this part, the main objective is to provide a detailed presentation of the validation and robustness of the proposed FZNN model (5) in solving TVLE (1) by means of a comparative simulation. Using a comparative simulation strategy, different types of noise and parameters are systematically modelled with the aim of comprehensively assessing the performance of the FZNN model (5). This process is not only used to validate the model, but also highlights the robustness of the FZNN model (5) under various noise and parameter conditions.

5.1 First illustrative example

In this example, several investigations are taken to demonstrate excellent performance of the proposed FZNN model (5) employing the below coefficients:

$$P(t) = \begin{bmatrix} \sin(2t) & \cos(2t) \\ -\cos(2t) & \sin(2t) \end{bmatrix}, \quad Q(t) = \begin{bmatrix} -\sin(2t) \\ -\cos(2t) \end{bmatrix}. \quad (14)$$

In the way, the following theoretical solution for $y(t)$ can be obtained:

$$y^*(t) = \begin{bmatrix} \cos(4t) \\ -\sin(4t) \end{bmatrix},$$

which is presented here to check the correctness of the proposed FZNN model (5). The randomly initial state $y(0) \in [-0.5; 0.5]$ and coefficient (14) for TVLE (1) are also employed.

(1) Noise-Free Condition: The process of solving the TVLE (1) in a noise-free environment using the FZNN model (5) is demonstrated by Fig. 1, which yields key observations.

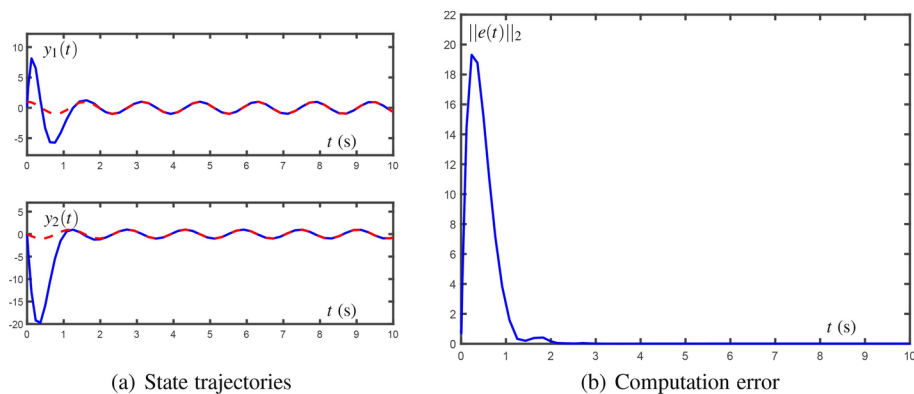


Fig. 1 Simulation results of FZNN model (5) with $\eta = 1$, $\gamma = 10$ under noiseless environment

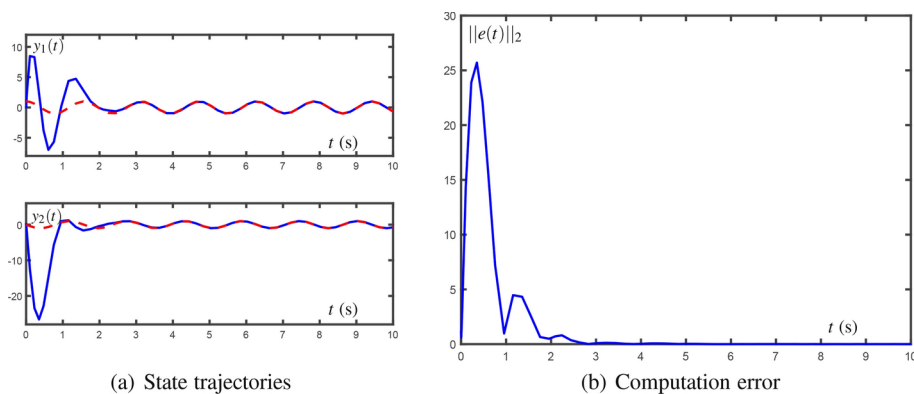


Fig. 2 Simulation results of FZNN model (5) with $\eta = 1$, $\gamma = 10$ under constant noise environment

The FZNN model (5) successfully obtains a theoretical solution to TVLE (1), providing intuitive evidence of the accuracy of the modified ZNN model. In the evaluation of the computational error, the FZNN model (5) exhibits a tendency to rapidly converge to zero within a short period of time, which implies that the results of the model are gradually approaching the theoretical solution during the computational process, proving the high accuracy of the model. That is, the fuzzy logic controller does not significantly affect the convergence properties of the ZNN model. It emphasizes the effectiveness and robustness of the FZNN model (5) in a noise-free environment. Overall, by analysing the theoretical solutions and computational errors of the FZNN model (5), the conclusion provides strong support for the feasibility of the model in this study.

- (2) Constant noise:** With the presentation in Fig. 2, the proposed FZNN model (5), where $\eta = 1$ and $\gamma = 10$, is used to process the TVLE (1) with a factor of (14) and observe its simulation results in a constant noise environment, i.e., $\delta_1(t) = [0.1; 0.1]$.

Figure 2 demonstrates that computational error gradually converges to zero as time progresses. This shows that despite the presence of error at the beginning of the simulation,

the FZNN model (5) has excellent convergence and adaptability in dealing with the TVLE (1), and is ultimately capable of yielding highly accurate solutions in noisy environment.

Aiming at verifying the effect of the parameter η on the convergence properties of the FZNN model (5), a parameter setting of $\eta = \gamma = 10$ is used in this study and applied to solve the TVLE (1) with the same coefficients (14) and constant noise as in Fig. 3. By observing the results in Fig. 3, the initial computational error with a larger value of η is smaller. This indicates that larger value of η helps to improve the accuracy of the model and reduce the discrepancy between the computed result and theoretical solution. Choosing an appropriate value of η affects the magnitude of the computational error of solving TVLEs. This provides useful guidance for further optimising the parameter selection of the FZNN model (5).

- (3) Harmonic Noise:** The excellent results of the FZNN model (5) in suppressing the low frequency harmonic noise $\delta_2(t) = [\cos(0.05t); \cos(0.05t)]$ are clearly presented by the demonstration in Fig. 4. In this simulation, the parameters were set as $\eta = \gamma = 10$ to successfully eliminate the effect of lower frequency harmonic noise on the solution of the TVLE (1) with factor (14).

The results reveal that the FZNN model (5) exhibits excellent performance in suppressing harmonic noise. It underlines the ability of the presented FZNN model (5) to efficiently suppress harmonic noise, providing substantial evidence of its feasibility in complex environments.

To further validate the robustness, simulations of higher-frequency harmonic noise $\delta_3(t) = [\cos(0.5t); \cos(0.5t)]$ are carried out using the IZNN (4) and the proposed FZNN (5) with the parameter setting $\eta = \gamma = 10$ and considering the coefficients of TVLE (1) as (14). Figures 5 and 6 correspond to the IZNN (4) and the FZNN (5), respectively. Error in Fig. 5b can not converge to and remain zero. That is, the IZNN (4) will fail with higher frequency harmonic noise. Figure 6 clearly demonstrates that the FZNN model (5) performs equally well in the higher frequency harmonic noise environment. The FZNN model (5) is still able to find its theoretical solution of the TVLE

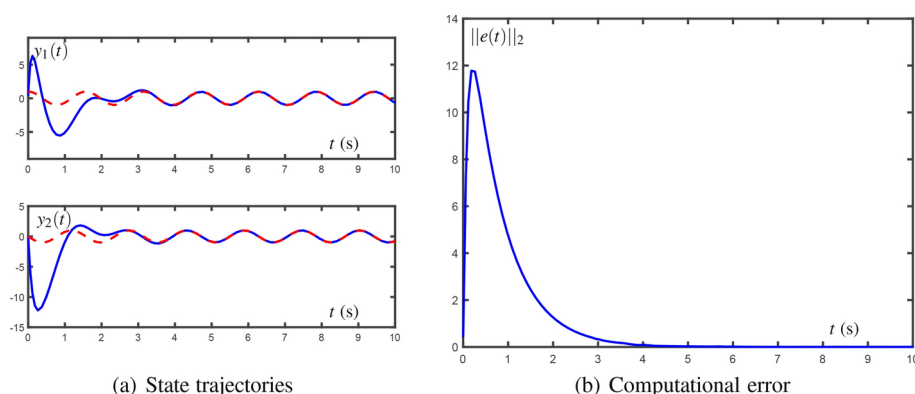


Fig. 3 Simulation results of FZNN model (5) with $\eta = \gamma = 10$ under constant noise environment

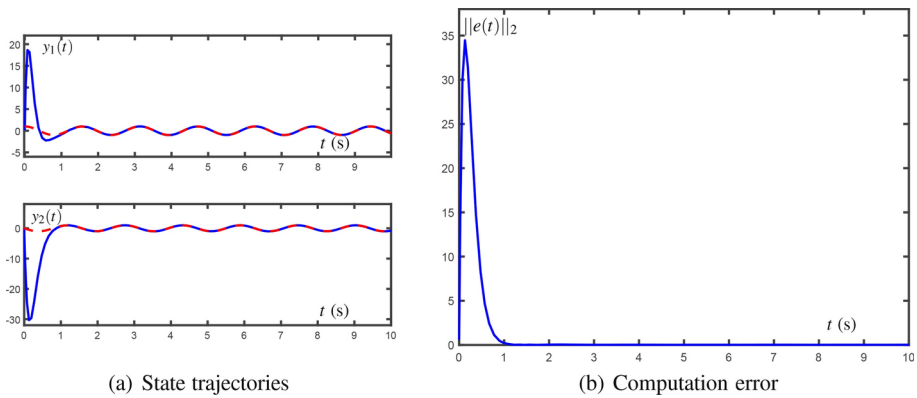


Fig. 4 Simulation results of FZNN model (5) with $\eta = \gamma = 10$ under low frequency harmonic noise environment

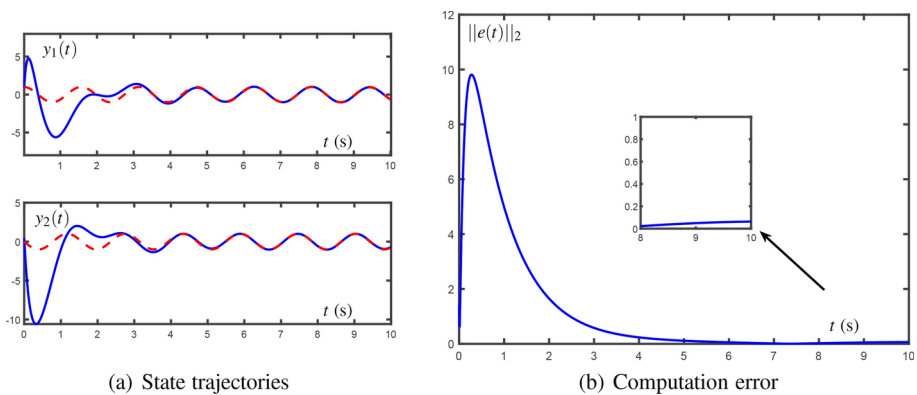


Fig. 5 Simulation results of utilizing IZNN model (4) with $\eta = \gamma = 10$ under high frequency harmonic noisy environment

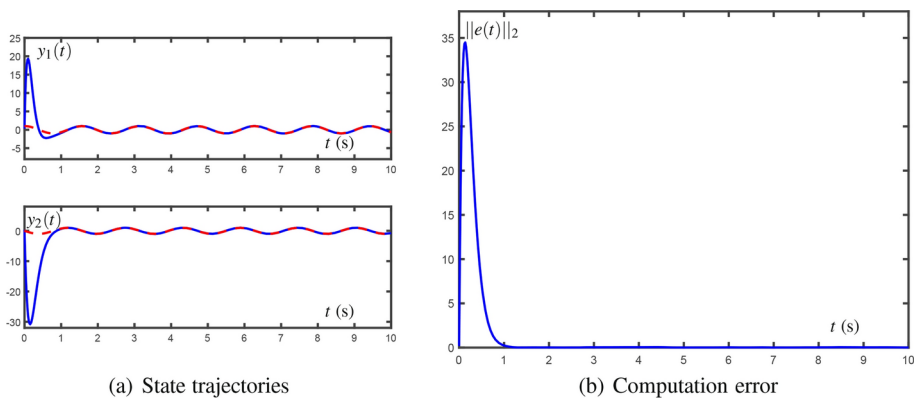


Fig. 6 Simulation results of utilizing proposed FZNN model (5) with $\eta = \gamma = 10$ under high frequency harmonic noisy environment

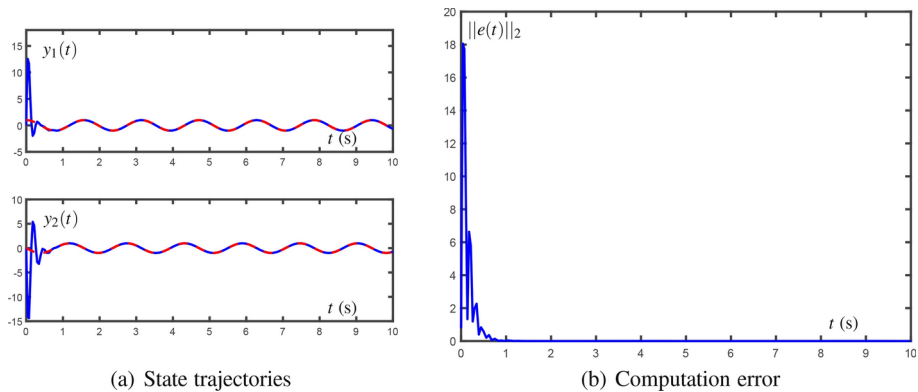


Fig. 7 Simulation results of utilizing proposed FZNN model (5) with $\eta = 10$, $\gamma = 100$ under high frequency harmonic noisy environment

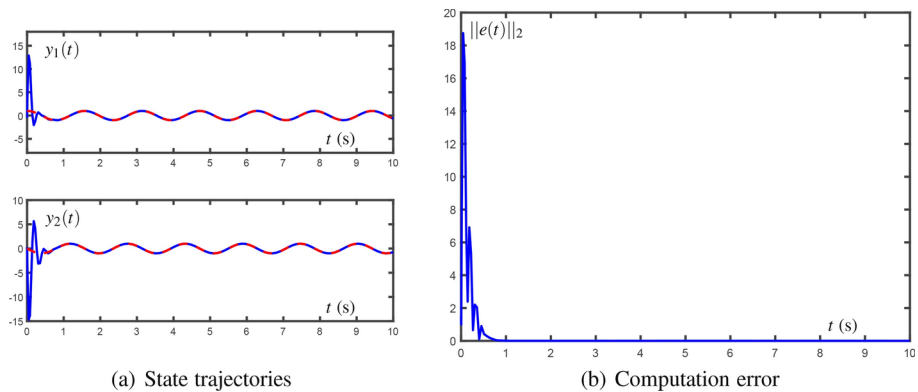


Fig. 8 Simulation results of utilizing proposed FZNN model (5) with $\eta = 10$, $\gamma = 100$ under smaller amplitude boundless noise environment

(1) efficiently under the interference of higher frequency harmonic noise, verifying the robustness of the FZNN model (5).

On the other hand, in order to highlight the effect of γ on the convergence rate of the FZNN model (5), a larger value of γ is adopted for the simulation. Figure 7 is the simulation results with $\eta = 10$, $\gamma = 100$, and the same coefficient (14) as well as harmonic noise as the Fig. 5. As can be observed from the results in Fig. 7, a larger value of γ indeed leads to a smaller initial computation error accelerates the adaptation process of the FZNN model (5) to the dynamic changes of the system.

(4) Boundless Noise: In an effort to prove the validity of the FZNN model (5) under different noises, it was examined in two boundless noise environment, i.e., $\delta_4(t) = [0.01t; 0.01t]$ and $\delta_5(t) = [0.1t; 0.1t]$.

Figures 8 and 9, corresponding to the boundless noise $\delta_4(t)$ and $\delta_5(t)$, respectively, show that the theoretical solution of the TVLE (1) can be obtained accurately and the

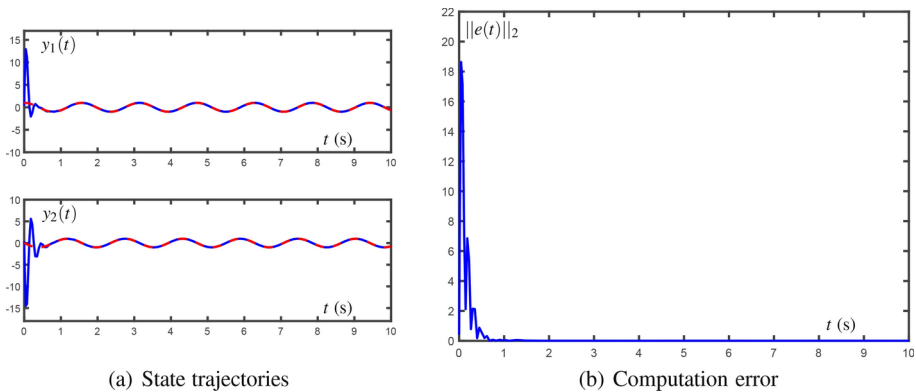


Fig. 9 Simulation results of utilizing proposed FZNN model (5) with $\eta = 10, \gamma = 100$ under larger amplitude boundless noise environment

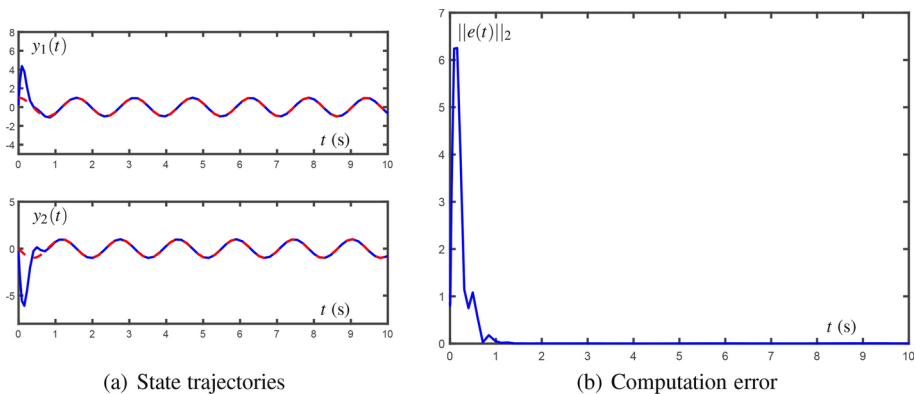


Fig. 10 Simulation results of utilizing proposed FZNN model (5) with $\eta = 100$ and $\gamma = 10$ under gaussian white noise environment

computational error converges to zero. It indicates the effectiveness of the FZNN model (5) in solving the TVLE (1) in boundless noise environments.

(5) Gaussian white noise: To further demonstrate the performance of the FZNN model (5) in different noise environment, Fig. 10 display the simulation result of solving the TVLE (1) in gaussian white noise $\delta_6(t) = [e^{0.1t}; e^{0.1t}]$. The corresponding result in Fig. 10 shows the convergence of state trajectories and computation error. This verifies the effectiveness of the FZNN model (5) in eliminating noise while solving the TVLE (1).

5.2 Second illustrative example

In this example, to demonstrate the validity of the FZNN model (5) more comprehensively, a six-dimensional coefficient matrix (15) is applied in this case in order to solve the TVLE (1), while noise-free and different noise scenarios (16–19) are considered

separately. And the each element of coefficient matrix (15) are $p_1(t) = 6 + \sin(3x)$ and $p_n(t) = \cos(3x)/(n-1)$, $n = 2, \dots, 6$. The randomly initial state $y(0) \in [-0.5; \dots; 0.5]^6$ and coefficient (15) are employed for the TVLE (1).

$$P(t) = \begin{bmatrix} p_1(t) & p_2(t) & \cdots & p_6(t) \\ p_2(t) & p_1(t) & \cdots & p_5(t) \\ \vdots & \vdots & \ddots & \vdots \\ p_6(t) & p_5(t) & \cdots & p_1(t) \end{bmatrix} \in R^{6 \times 6}, \quad \text{and} \quad Q(t) = \begin{bmatrix} \cos(3t) \\ 2 - \cos(3t) \\ \sin(3t) \\ \cos(3t) \\ 1 + \sin(3t) \\ \sin(3t) \end{bmatrix} \in R^6, \quad (15)$$

$$\delta_7(t) = [0.1; \dots; 0.1] \in R^6, \quad (16)$$

$$\delta_8(t) = [\cos(0.05t); \dots; \cos(0.05t)] \in R^6, \quad (17)$$

$$\delta_9(t) = [0.1t; \dots; 0.1t] \in R^6. \quad (18)$$

$$\delta_{10}(t) = [e^{0.1t}; \dots; e^{0.1t}] \in R^6. \quad (19)$$

Figures 11–14 present the simulation results, corresponding respectively to the different noise cases (16)–(19). Evidently, in all four figures, the excellent performance of the FZNN model (5) in addressing the TVLE (1) with a six-dimensional coefficient matrix in either noise-free or different noise environment is clearly demonstrated. The model not only accurately obtains the theoretical solution of the TVLE (1), but also the computational error consistently tends to zero throughout the simulations.

Overall, the superior properties (including adaptability, robustness and noise resistance) shown in solving the TVLE (1) are theoretically analyzed and simulated.

Remark From all the above simulation results Figs. 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, and 14, there are some important conclusions can be exacted. In terms of model validation, the adaptability and robustness of the FZNN model (5) to different noises and parameters

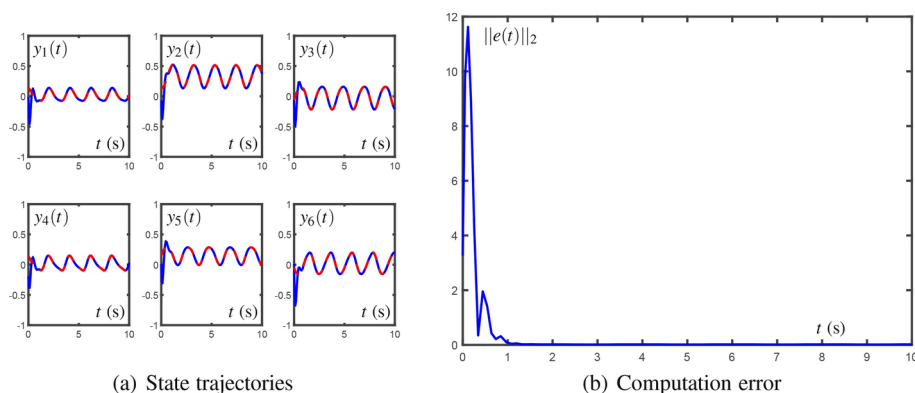


Fig. 11 Simulation results of utilizing proposed FZNN model (5) with $\eta = 100$ and $\gamma = 10$ under constant noisy environment

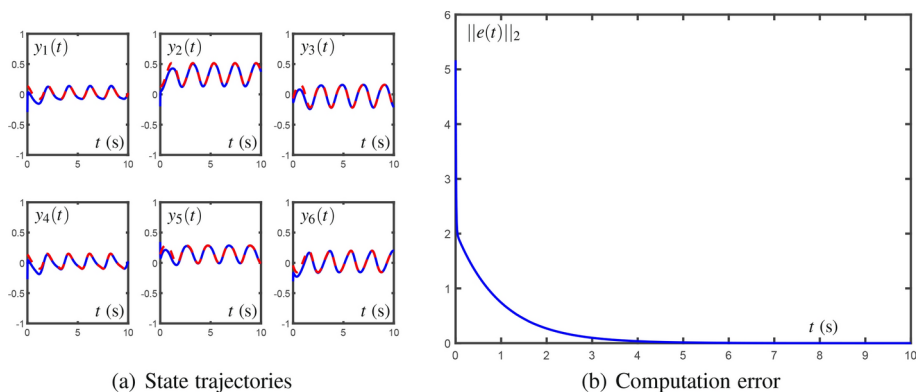


Fig. 12 Simulation results of utilizing proposed FZNN model (5) with $\eta = 100$ and $\gamma = 10$ under harmonic noisy environment

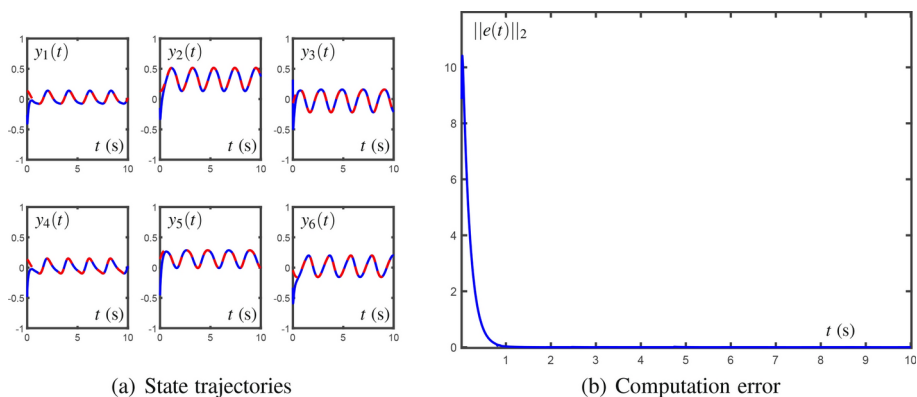


Fig. 13 Simulation results of utilizing proposed FZNN model (5) with $\eta = 100$ and $\gamma = 10$ under boundlessly noisy environment

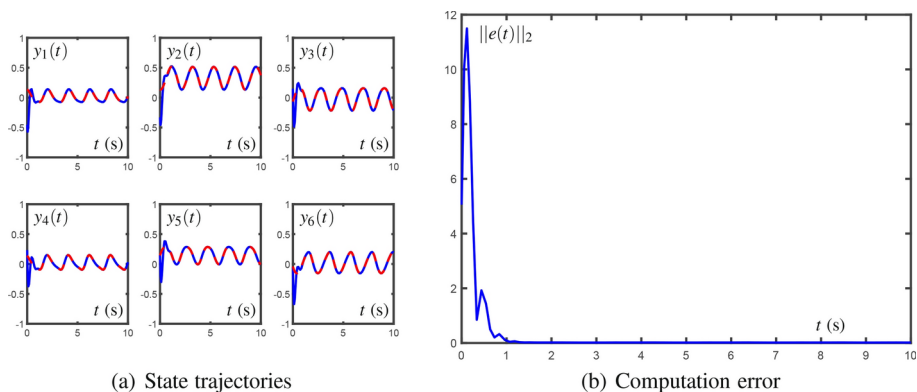


Fig. 14 Simulation results of utilizing proposed FZNN model (5) with $\eta = 100$ and $\gamma = 10$ under gaussian white noisy environment

are successfully verified through comparative simulations. The simulation results clearly demonstrate that the FZNN model (5) is able to find the theoretical solution of the TVLE (1) effectively under different conditions and the computational error consistently converges to zero throughout the simulations. Further parameter sensitivity validation shows that the model is significantly sensitive to the parameters. Larger values of η and γ lead to smaller initial computational errors, further highlighting the importance of the FZNN model (5) in parameter selection. Finally the noise environment, the simulation results demonstrate the effectiveness and robustness of the FZNN model (5) under different noises. The FZNN model (5) is still able to accurately obtain the TVLE (1) solutions in complex noise environments, proving its robustness in practical applications.

6 Conclusion

As the complexity of a system increases, the available accurate information decreases, leading to controlling accurately the system more difficultly. To address the challenge, this paper proposes a novel fuzzy control strategy, the FZNN model (5), based on noise-resistant zeroing neural network model. The innovation of the FZNN model (5) is its ability to adaptively scale the parameters to cope with different noise environments faced by the system through real-time monitoring of the feedback computational errors. It makes the model more adaptive and robust as well as improves its practical applicability in complex systems. The simulation results fully confirm the design expectation of the model and verify its excellent performance in noisy environments.

Although the proposed FZNN model (5) exhibits excellent performances when involved in noisy or uncertain situations, there are still some disadvantages. For example, responsiveness of fuzzy controllers may be limited since they rely on fuzzy control rules and reasoning process. In the future, we expect to design more superior fuzzy controllers to speed the decision. And we will also focus on further exploring the potential value of the FZNN model (5) for a broader range of application scenarios in future research. Due to the high adaptability of the model, we expect to leverage its superiority in a variety of real-world applications to provide more flexible and efficient solutions in the field of complex system control. This study provides an innovative idea to address the challenges of increasing system complexity and nonlinearity, also provides a useful reference for future in-depth research in related fields.

Appendix

In this appendix, the derivation of the OZNN model (2) is presented for better understanding.

Regarding the TVLE (1), the vector-valued error function is defined as follows:

$$e(t) = P(t)y(t) - Q(t) \in R^n.$$

To guarantee each element $e_i(t)$ of $e(t)$ converges to zero, the following design formula is utilized (Zhang and Li 2009):

$$\frac{de(t)}{dt} = -\eta\phi(e(t)), \quad (20)$$

with γ being the design parameter, and $\phi(\cdot)$ being the monotonically increasing odd activation function.

Substantiating $e(t) = P(t)y(t) - Q(t)$ into (20) yields

$$\dot{P}(t)y(t) + P(t)\dot{y}(t) - \dot{Q}(t) = \gamma\phi(P(t)y(t) - Q(t)),$$

which is further reformulated as follows:

$$P(t)\dot{y}(t) = -\dot{P}(t)y(t) + \dot{Q}(t) - \gamma\phi(P(t)y(t) - Q(t)). \quad (21)$$

Notably, (21) is exactly the OZNN model (2) designed by Zhang and Li for solving the time-varying linear equation (1) (Zhang and Li 2009).

Acknowledgements The authors would like to thank the editors and reviewers for their time and effort in evaluating this paper and for the constructive comments for the improvement of its presentation and quality.

Author contributions All authors contributed to the study conception and design. Material preparation, data collection and analysis were performed by Dongsheng Guo, Chan Zhang, Naimeng Cang. The first draft of the manuscript was written by Dongsheng Guo and Chang Zhang. All authors commented on previous versions of the manuscript. All authors read and approved the final manuscript.

Funding This paper is supported by the Scientific Research Fund of Hainan University under Grant (KYQD(ZR)23025).

Data availability The code or data were available in this manuscript.

Declarations

Conflict of interest The authors have no relevant financial or non-financial interests to disclose.

Ethical approval This paper does not contain any studies with human participants or animals performed by any of the authors.

Consent to participate Informed consent was obtained from all individual participants included in the study.

Consent to publish The authors affirm that human research participants provided informed consent for publication of the images in all figures.

Open Access This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by-nc-nd/4.0/>.

References

- Byrd RH, Waltz RA (2011) An active-set algorithm for nonlinear programming using parametric linear programming. *Optim Methods Soft* 26(1):47–66
- Cao Y (2012) Aggregating multiple classification results using Choquet integral for financial distress early warning. *Expert Syst Appl* 39(2):1830–1836
- Chen Y, Yi C, Qiao D (2013) Improved neural solution for the Lyapunov matrix equation based on gradient search. *Inf Process Lett* 113(22–24):876–881
- Ding L, Wang L, Liu S (2023) Recurrent graph encoder for syntax-aware neural machine translation. *Int J Mach Learn Cybern* 14(4):1053–1062
- Frinken V, Uchida S (2015) Deep BLSTM neural networks for unconstrained continuous handwritten text recognition. In: 2015 13th international conference on document analysis and recognition (ICDAR). IEEE, pp 911–915
- Hopfield J, Tank D (1985) “Neural” computation of decisions in optimization problems. *Biol Cybern* 52(3):141–152
- Jia L, Xiao L, Dai J, Wang Y (2023) Intensive noise-tolerant zeroing neural network based on a novel fuzzy control approach[J]. *IEEE Trans Fuzzy Syst* 31(12):4350–4360
- Jin L, Zhang Y, Li S, Zhang Y (2016) Modified ZNN for time-varying quadratic programming with inherent tolerance to noises and its application to kinematic redundancy resolution of robot manipulators. *IEEE Trans Ind Electron* 63(11):6978–6988
- Lv X, Xiao L, Tan Z (2019) Improved Zhang neural network with finite-time convergence for time-varying linear system of equations solving. *Inf Process Lett* 147:88–93
- Oppenheim A, Willsky A, Nawab S (1997) Signals and systems. Prentice-Hall, NJ, USA
- Patrick D, Alle-Jan V (1998) Time-varying systems and computations. Springer, Boston
- Precup R, Hellendoorn H (2011) A survey on industrial applications of fuzzy control. *Comput Ind* 62(3):213–226
- Qiu B, Zhang Y, Yang Z (2018) New discrete-time ZNN models for least-squares solution of dynamic linear equation system with time-varying rank-deficient coefficient. *IEEE Trans Neural Netw Learning Syst* 29(11):5767–5776
- Roy S, Bhaumik A (2018) Intelligent water management: a triangular type-2 intuitionistic fuzzy matrix games approach. *Water Resour Manage* 32:949–968
- Shu O, Eddie N, Ru S, Rajendra A (2021) Automated diagnosis of arrhythmia using combination of CNN and LSTM techniques with variable length heart beats. *Comput Biol Med* 102:278–287
- Supratak A, Dong H, Wu C, Wu Y (2017) DeepSleepNet: a model for automatic sleep stage scoring based on raw single-channel EEG. *IEEE Trans Neural Syst Rehabil Eng* 25(11):1998–2008
- Tank D, Hopfield J (1986) Simple ‘neural’ optimization networks: an A/D converter, signal decision circuit, and a linear programming circuit. *IEEE Trans Circuits Syst* 33(5):533–541
- Xiao L, Li S, Li K, Liao B (2018) Co-design of finite-time convergence and noise suppression: a unified neural model for time varying linear equations with robotic applications. *IEEE Trans Syst Man Cybern: Syst* 50(12):5233–5243
- Xiao L, Cao Y, Dai J, Jia L, Tan H (2020) Finite-time and predefined-time convergence design for zeroing neural network: theorem, method, and verification. *IEEE Trans Ind Inf* 17(7):4724–4732
- Xie Z, Schwartz O, Prasad A (2018) Decoding of finger trajectory from ECoG using deep learning. *J Neural Eng* 15(3):036009
- Yi C, Chen Y, Lu Z (2011) Improved gradient-based neural networks for online solution of Lyapunov matrix equation. *Inf Process Lett* 111(16):780–786
- Zadeh H (1965) Fuzzy sets. *Inf Control* 8(3):338–353
- Zhang Y, Guo D (2015) Zhang functions and various models. Springer, Heidelberg
- Zhang Y, Li Z (2009) Zhang neural network for online solution of time-varying convex quadratic program subject to time-varying linear-equality constraints. *Phys Lett A* 373(18–19):1639–1643
- Zhang Z, Yan Z (2018) Hybrid-level joint-drift-free scheme of redundant robot manipulators synthesized by a varying-parameter recurrent neural network. *IEEE Access* 6:34967–34975
- Zhang Z, Yan Z (2019) An adaptive fuzzy recurrent neural network for solving the nonrepetitive motion problem of redundant robot manipulators. *IEEE Trans Fuzzy Syst* 28(4):684–691
- Zhang Y, Yi C (2011) Zhang neural networks and neural-dynamic method. Nova, New York
- Zhang Y, Jiang D, Wang J (2002) A recurrent neural network for solving Sylvester equation with time-varying coefficients. *IEEE Trans Neural Netw* 13(5):1053–1063
- Zhang Y, Chen K, Li X, Yi C, Zhu H (2008) Simulink modeling and comparison of Zhang neural networks and gradient neural networks for time-varying Lyapunov equation solving. In: Fourth International Conference on Natural Computation. IEEE, pp 521–525

- Zhang Y, Chen K, Tan HZ (2009a) Performance analysis of gradient neural network exploited for online time-varying matrix inversion. *IEEE Trans Autom Control* 54(8):1940–1945
- Zhang Y, Chen Z, Chen K (2009b) Convergence properties analysis of gradient neural network for solving online linear equations. *Acta Autom Sin* 35(8):1136–1139
- Zhang Y, Shi Y, Yang Y, Ke Z (2010) Convergence analysis of Zhang neural networks solving time-varying linear equations but without using time-derivative information. In: *IEEE International Conference on Control and Automation (ICCA)*. IEEE, PP 1215–1220
- Zhang Y, Yi C, Guo D, Zheng J (2011a) Comparison on Zhang neural dynamics and gradient-based neural dynamics for online solution of nonlinear time-varying equation. *Neural Comput Appl* 20:1–7
- Zhang Y, Mu B, Zheng H (2011b) Discrete-time Zhang neural network and numerical algorithm for time-varying linear equations solving. In: *2011 IEEE International Conference on Mechatronics and Automation*. IEEE, pp 938–943
- Zhang Y, Yu X, Mu B, Fan Z, Zheng H (2013) New discrete-time ZNN models and numerical algorithms derived from a new Zhang function for time-varying linear equations solving. In: *2013 Ninth International Conference on Natural Computation (ICNC)*. IEEE, PP 222–226
- Zhang Z, Yan Z, Fu T (2018) Varying-parameter RNN activated by finite-time functions for solving joint-drift problems of redundant robot manipulators. *IEEE Trans Ind Inf* 14(12):5359–5367
- Zhang Y, Qi Z, Qiu B, Yang M, Xiao M (2019) Zeroing neural dynamics and models for various time-varying problems solving with ZLSF models as minimization-type and Euler-type special cases. *IEEE Comput Intell Mag* 14(3):52–60
- Zhang Z, Lin W, Zheng L, Zhang P, Qu X, Feng Y (2020) A power-type varying gain discrete-time recurrent neural network for solving time-varying linear system. *Neurocomputing* 388:24–33

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.